

MSE 317

Ceramic Materials

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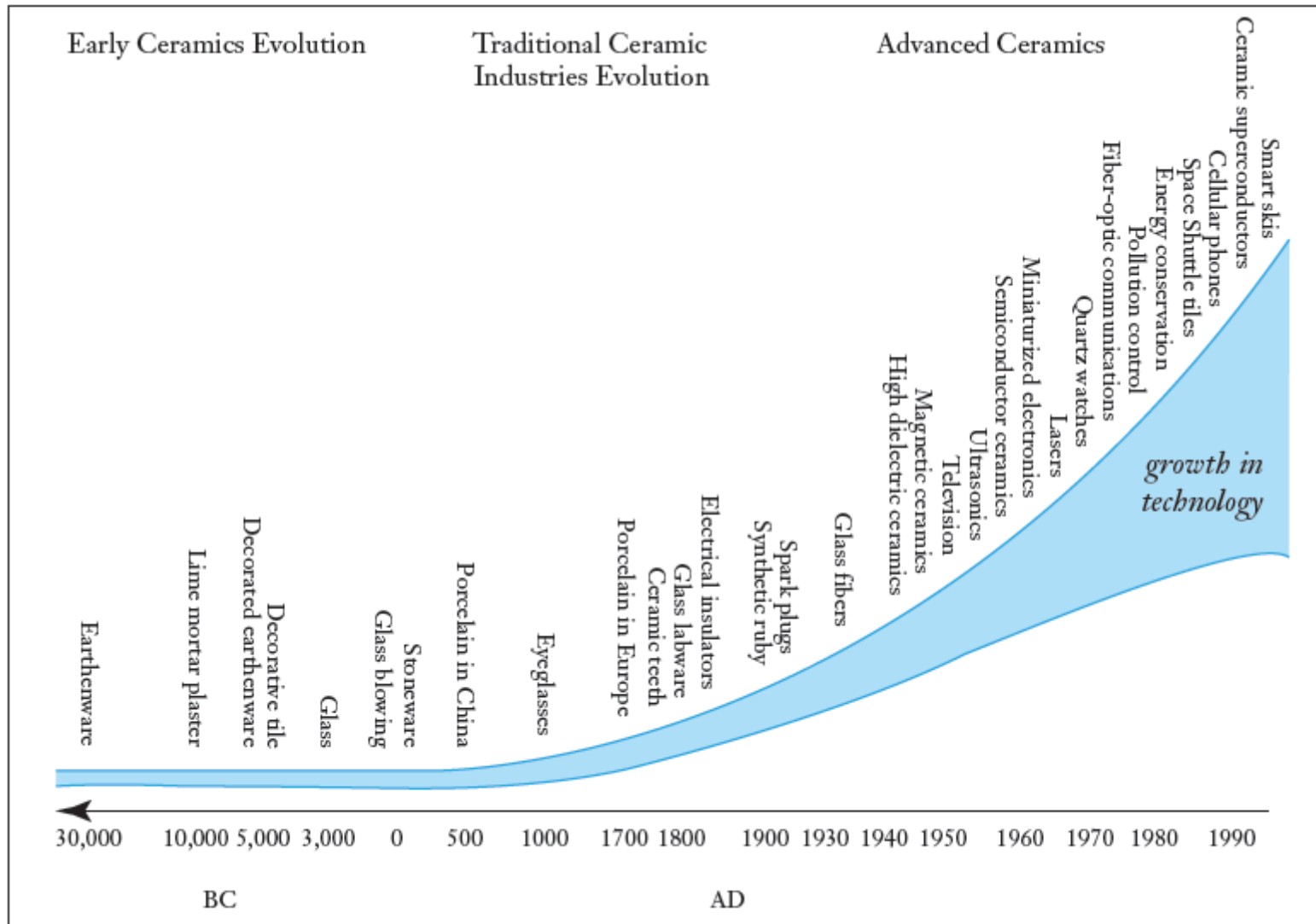
Definitions

Solid Materials are often divided into three distinct Class:

1. Metals: atoms held together by delocalized electrons that overcome the mutual repulsion between the ions cores;
2. Polymers: Macromolecules formed by covalent bonding of many simple molecular units. (C, H,O..)
3. **Ceramics**: with mixed bonding-a combination of covalent, ionic, and metallic.

‘A ceramic is nonmetallic, inorganic solids.’ by Kingery

The historical evolution of ceramics



Ceramic Bonding

- Bonding:
 - Mostly ionic, some covalent.
 - Ionic character increases with difference in electronegativity.
- Large vs small ionic bond character:

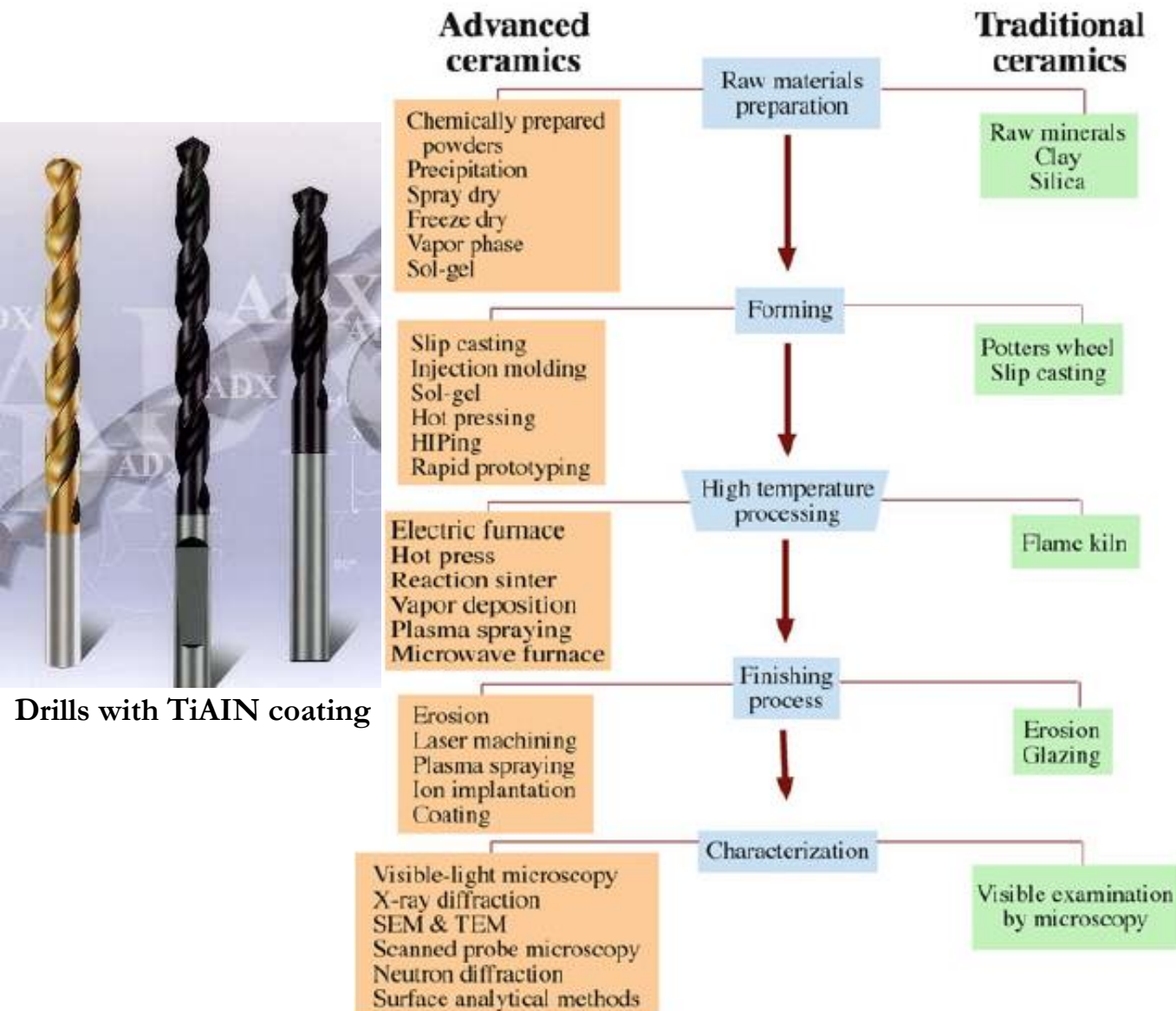
IA																	0
H																	He
2.1	IIA											IIIA	IVA	VA	VIA	VIIA	-
Li	Be											B	C	N	O	F	Ne
1.0	1.5											2.0	2.5	3.0	3.5	4.0	-
Na	Mg											Al	Si	P	S	Cl	Ar
0.9	1.2											1.5	1.8	2.1	2.5	3.0	-
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
0.8	1.0	1.3	1.5	1.6	1.6	1.5	1.8	1.8	1.8	1.9	1.6	1.6	1.8	2.0	2.4	2.8	-
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
0.8	1.0	1.2	1.4	1.6	1.8	1.9	2.2	2.2	2.2	1.9	1.7	1.7	1.8	1.9	2.1	2.5	-
Cs	Ba	La-Lu	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
0.7	0.9	1.1-1.2	1.3	1.5	1.7	1.9	2.2	2.2	2.2	2.4	1.9	1.8	1.8	1.9	2.0	2.2	-
Fr	Ra	Ac-No															
0.7	0.9	1.1-1.7															

CaF₂: large

SiC: small

Adapted from Fig. 2.7, *Callister 7e*. (Fig. 2.7 is adapted from Linus Pauling, *The Nature of the Chemical Bond*, 3rd edition, Copyright 1939 and 1940, 3rd edition. Copyright 1960 by Cornell University.

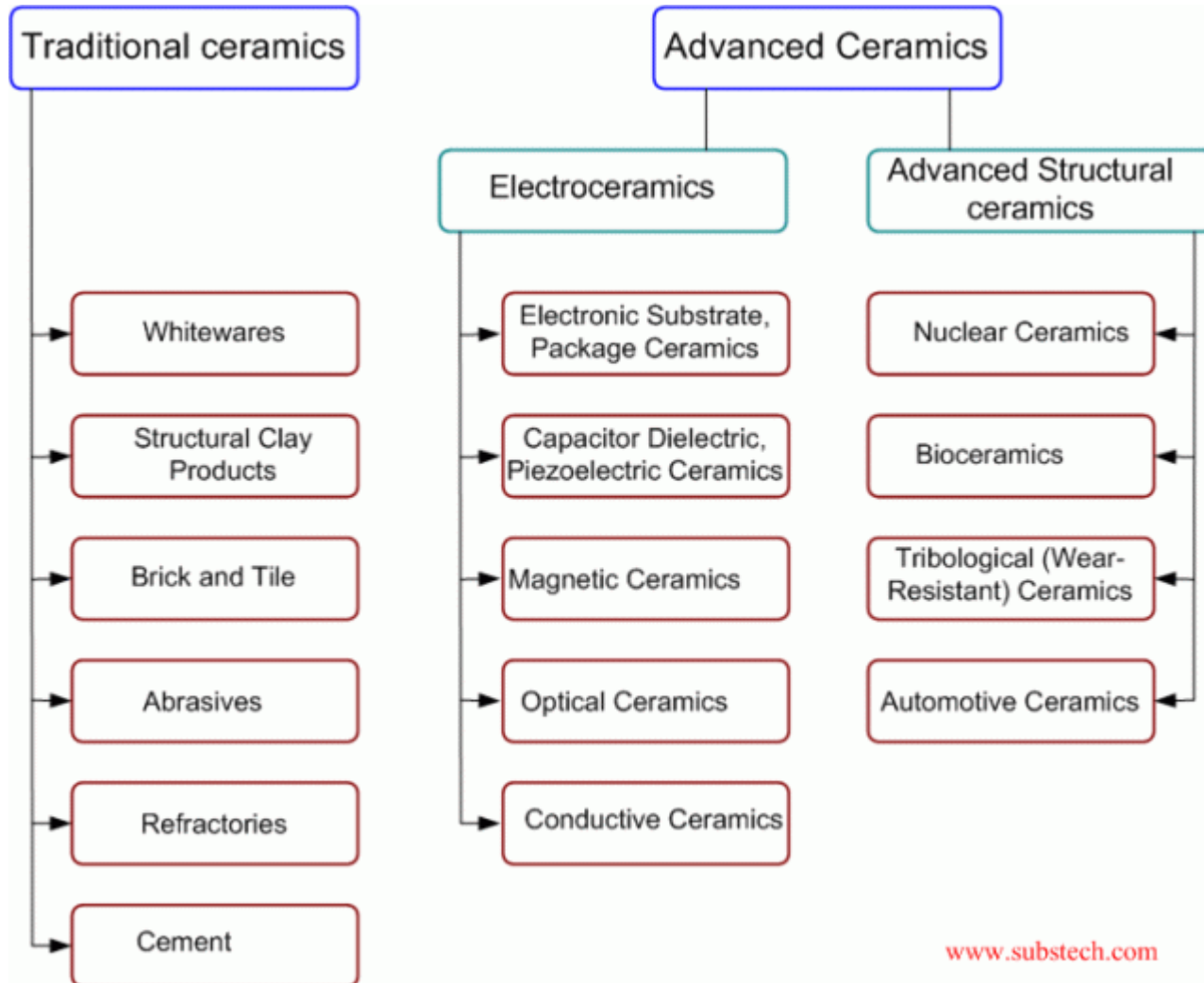
Classification of Ceramics



Drills with TiAlN coating



Application base Classification of Ceramic Materials



Characteristics base Classification of Ceramic Materials

- **Structural Ceramics** (Mechanical properties: strength, toughness, hardness, creep resistance)



Two Kyocera ceramic knives (ZrO_2)



Ceramic Si_3N_4 bearing parts



The Porsche Carrera GT's silicon carbide disk brake



High Mechanical Performances;
Used at high temperatures;
Or combine both above



Radial rotor made from Si_3N_4 for a gas turbine engine

- **Functional Ceramics** (electric, magnetic, optical properties)

Functional Ceramics

Functionality	Material	Applications
Resistors	SiC, MoS ₂ , LaCrO ₃	Heating elements for high temperature furnaces
Thermistors (NTCR & PTCR)	Spinel BaTiO ₃	Temperature sensors, self-regulating heating elements
Dielectrics with very low losses ($\epsilon_r = 3 - 10$)	Al ₂ O ₃ , AlN, cordierite	Substrates for electronic circuits and chip packaging
Dielectrics for microwave applications ($\epsilon_r = 30 - 80$)	BaTi ₄ O ₉ , Zr(Ti,Sn)O ₄ , BaMg _{1/3} Ta _{2/3} O ₃ , (Ba,Sr)TiO ₃ ,	MW resonators, filters and antennas for mobile communications and GPS devices, tunable MW devices
Temperature stable dielectrics ($\epsilon_r \approx 100$)	CaTiO ₃ , BaO-Nd ₂ O ₃ -TiO ₂	Capacitors with temperature-independent capacitance
Dielectrics with very high dielectric constant ($\epsilon_r \approx 3000$)	BaTiO ₃	Multilayer ceramic capacitors
Piezoelectric ceramics	Pb(Zr,Ti)O ₃ (PZT)	Transducers, actuators and resonators
Pyroelectric ceramics	Pb(Zr,Ti)O ₃	IR radiation detection and imaging

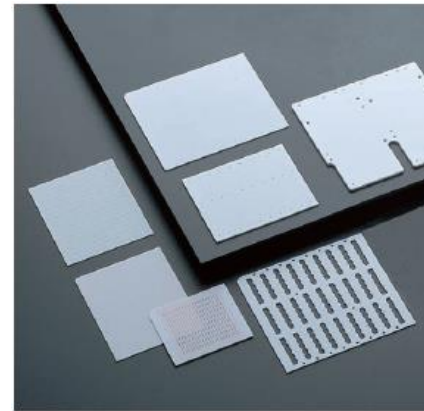
Functional Ceramics

Functionality	Material	Applications
Ferroelectric ceramics	$\text{Pb}(\text{Zr,Ti})\text{O}_3$ $\text{SrBi}_2\text{Ta}_2\text{O}_9$	Ferroelectric memories (FeRAMs)
Electrostrictive ceramics	$\text{PbMg}_{1/3}\text{Nb}_{2/3}\text{O}_3$ - PbTiO_3 (PMN-PT)	Actuators
Magnetic ceramics	Spinels $(\text{Ni,Zn})\text{Fe}_2\text{O}_4$ $\text{BaFe}_{12}\text{O}_{19}$ $\text{Y}_3\text{Fe}_5\text{O}_{12}$ (YIG)	Inductors Permanent magnets Microwave devices (radars)
Ionic conductors	Y:ZrO_2 (YSZ) Gd:CeO_2 β -alumina	Electrolytes for solid-oxide fuel cells (SOFCs), oxygen sensors Na-Batteries
Superconductors	$\text{YBa}_2\text{Cu}_3\text{O}_{7-x}$ (YBCO) MgB_2	Superconducting cables for magnets
Transparent ceramics	Al_2O_3 , MgAl_2O_4 , $\text{Y}_3\text{Al}_5\text{O}_{12}$ (YAG)	Phosphors, optical materials for lenses and laser systems, nose cones for heat-seeking missiles, high-pressure sodium street lamps
Optoelectronic materials	LiNbO_3 PLZT	Waveguides, frequency doublers, voltage-controlled optical switches, modulators

Functional Ceramics



Pressed and extruded parts (alumina, mullite, zirconia)



Thick (left) and thick (right) substrates (alumina)

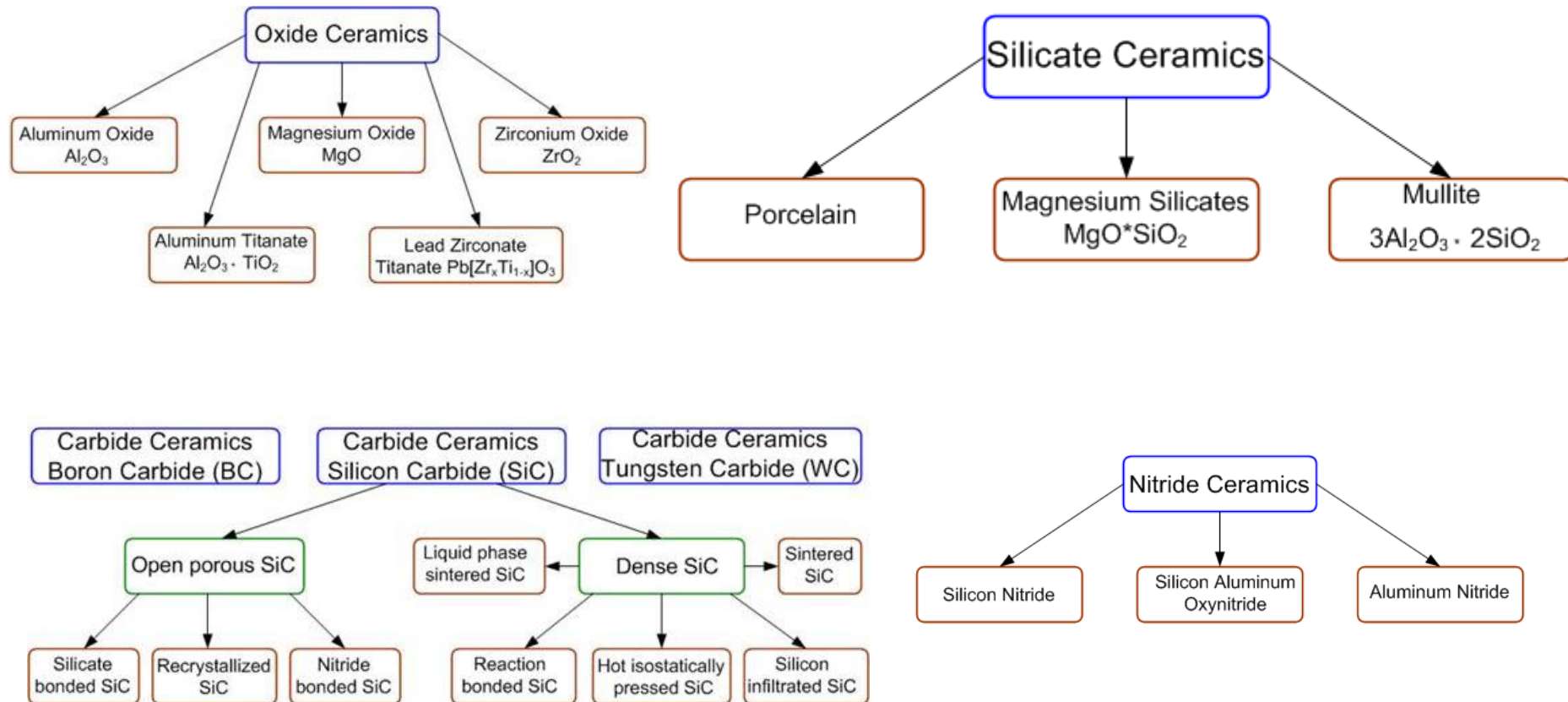


Ferrites cores



Microwave dielectric components

Composition base Classification of Ceramic Materials



Metals vs Ceramics

Metals	Ceramics
Crystal structure	Crystal structure
Large number of free electrons	Captive electrons
Metallic bond	Ionic/covalent bonds
Good electrical conductivity	Poor conductivity
Opaque	Transparent (in thin sections)
Uniform atoms	Different-size atoms
High tensile strength	Poor tensile strength ^a
Low shear strength	High shear strength
Good ductility	Poor ductility (brittle)
Plastic flow	None
Impact strength	Poor impact strength
Relatively high weight	Lower weight
Moderate hardness	Extreme hardness
Nonporous	Initial high porosity
High density	Initial low density

General Properties of Ceramic Materials

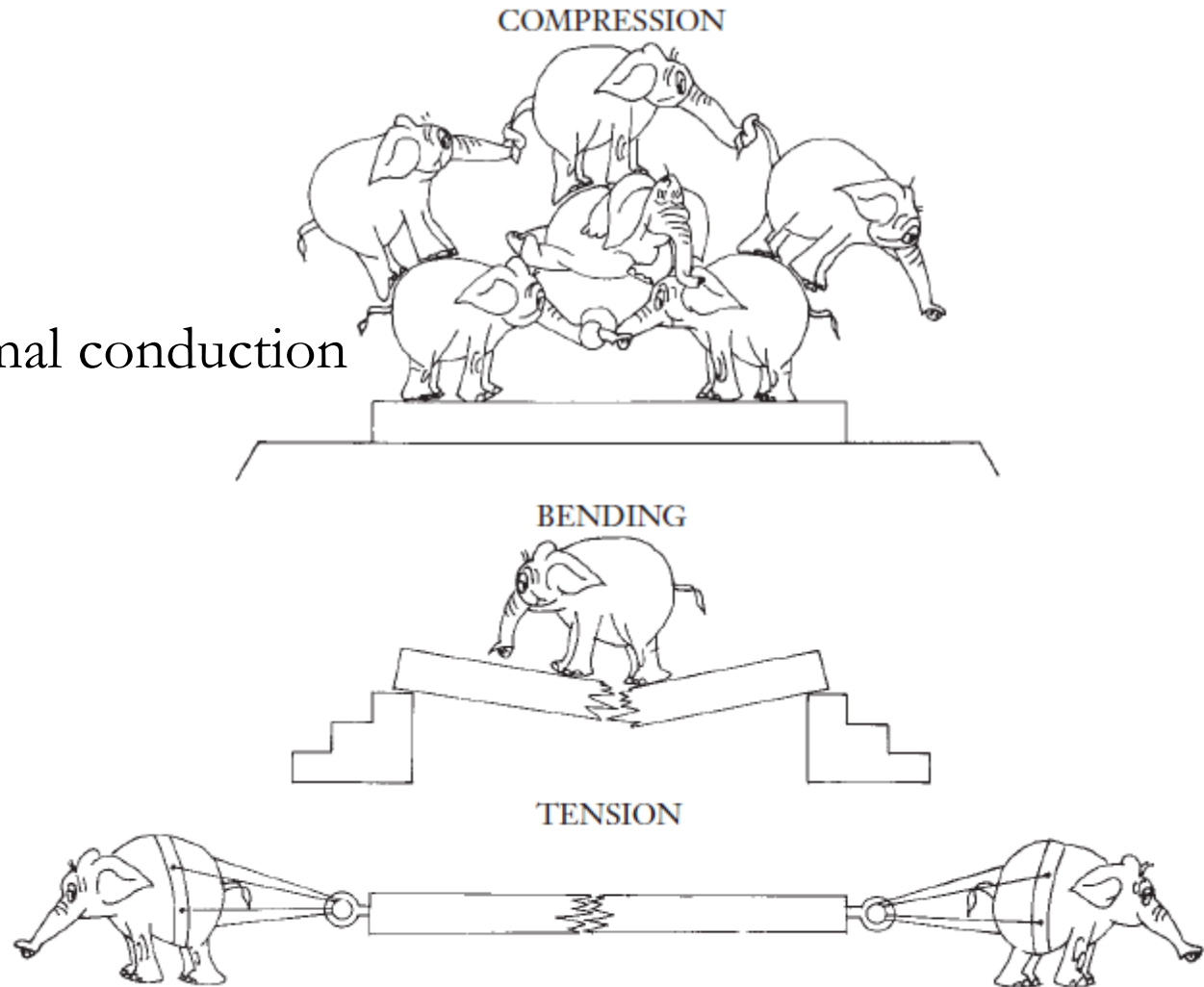
Brittleness;

Compression Strength;

Poor electrical and thermal conduction

Chemical Insensitivity;

Transparent



What will be Covered in this Course ?

